



University of Vavuniya

Third Examination in Information Technology - 2023

Second Semester - July 2025

IT3223 Advanced Database Management Systems (Theory)

Answer *Four* Questions Only

Time Allowed : Two hours

-
1. (a) Define each of the following in the context of database file storage and indexing: [20%]
- | | |
|-------------------|---------------------|
| i. Buffer | ii. Blocking factor |
| iii. Sparse index | iv. Dense index |
- (b) Describe *five* techniques which make data access more efficient on disk. [20%]
- (c) Accessing data from a disk block is expensive. Discuss the reasons based on the time components involved in accessing a disk block. [15%]
- (d) Consider a disk block with block size $B = 512$ bytes. A block pointer is $P = 6$ bytes long. A file has $r = 30,000$ Student records of fixed length. Each record has the following fields: Name (30 bytes), NIC_No(9 bytes), Reg_No (9 bytes), Address (40 bytes), Phone_No (10 bytes), Birth_date (8 bytes), Sex (1 byte), Department (4 bytes), and Faculty (4 bytes). An additional byte is used as a deletion marker.
- i. Calculate the record size in bytes, the blocking factor, and the number of file blocks, assuming an unspanned organization. [15%]

[This question is continued on the next page]

- ii. Assume that the file is ordered by the key field *NIC_No* and you need to construct a sparse primary index on *NIC_No*. Calculate the following using the given information above:
- A. The **index record length** and **index blocking factor**. [10%]
 - B. The **number of index entries** and the **required blocks** to store the primary index. [10%]
 - C. The **number of block accesses** needed to search for and retrieve a record from the file using the **primary index**. [10%]
2. (a) Describe the steps followed by a DBMS to **process a high-level query**. [20%]
- (b) Discuss the *five* types of cost considered during **cost-based query optimization**. [15%]
- (c) Consider the following description of a **Hospital** database:
- A hospital maintains a centralised system to manage the personal details of patients and staff, as well as information about hospital departments and treatment information. For each patient, the system records: *Patient_ID*, *Full_Name*, *Home_Address*, *Date_of_Birth*, and *Emergency_Contact_Number*. For each staff member, the system stores: *Staff_ID*, *First_Name*, *Last_Name*, *Job_Title* (e.g., Surgeon, Nurse, General Practitioner), and *Department_Name*. For each department, the system maintains: *Department_Name*, *Name of the Department Head*, *Location*, and *Number of beds available*. For treatment information, the system keeps details such as *Treatment_ID*, *Patient_ID*, *Staff_ID* and *Date_of_Treatment*.
- Assume that you have been requested to execute the query given below:
- List the names of all patients treated by ‘‘Surgeons’’ in departments that have more than 20 beds. Also, show the name of the department and the last name of the treating staff member.
- i. Write the above query in **SQL**. [10%]

[This question is continued on the next page]

4. (a) Describe briefly each of the following in the context of **locking protocols**: [20%]
- i. Shared lock
 - ii. Exclusive lock
 - iii. Expanding Phase
 - iv. Shrinking Phase
- (b) Discuss how **deadlock and starvation** problems can be resolved while applying **locking protocols**. [20%]
- (c) Explain the benefits of the **timestamp ordering** over the **locking** while implementing **concurrency control**. [20%]
- (d) Describe each of the following in the context of **in-place updating**:
- i. Write-Ahead Logging
 - ii. Steal/ No steal approach
 - iii. Force/ No force approach
- (e) Determine whether each of the following schedules is recoverable, cascading, non-recoverable, or cascadeless. Provide appropriate justification for your answer:
- i. $w_1(X); r_2(X); w_2(Y); c_1; r_2(Z); w_2(Z); c_2;$
 - ii. $w_1(X); r_2(X); w_2(Y); c_2; r_1(Z); w_1(Z); a_1;$
 - iii. $w_1(X); r_2(X); r_3(X); w_2(Y); a_1; a_2; a_3;$
 - iv. $w_1(X); c_1; r_2(X); w_2(Y); r_2(Z); w_2(Z); c_2;$ [20%]
5. (a) Describe *five* benefits of **distributed database systems** over **centralized database systems**. [20%]
- (b) Identify the **transparency mechanism** applied for each of the following situations:
- i. A customer logs into their online banking account while travelling abroad. They can access their account details seamlessly, even though the data is physically located in their home country's data center.
 - ii. A stock trading app allows users to view real-time share prices. These prices are replicated across multiple servers for load balancing.

[This question is continued on the next page]

- iii. A telecom company stores billing data separately from usage details for optimisation.
- iv. A government tax portal upgrades its schema and storage engine, switching from SQL to a distributed NoSQL backend. Tax officers continue using the same interface and queries.
- v. A customer submits a loan application on a banking portal. The system runs distributed queries, updates multiple tables, and verifies credit scores from external services.

[25%]

(c) Consider the relation **Account_Details** given by Table 1:

Account_ID	Account_Holder_Name	Account_Type	Balance	Branch_Code
A1001	Priya	Savings	52,000.00	BR001
A1002	Rohan	Current	125,000.00	BR002
A1003	Meena	Fixed Deposit	500,000.00	BR001
A1004	John	Savings	38,500.00	BR003
A1005	Ayesha	Current	98,000.00	BR004
A1006	Kavya	Savings	12,300.00	BR002
A1007	Arjun	Fixed Deposit	250,000.00	BR001
A1008	Nimal	Current	67,700.00	BR005
A1009	Lakshmi	Savings	44,200.00	BR003
A1010	Harish	Fixed Deposit	300,000.00	BR002

Table 1: Account_Details

- i. Describe briefly each of the following types of **fragmentation techniques** used by the **distributed database systems**:
 - A. Horizontal fragmentation
 - B. Vertical fragmentation
 - C. Hybrid fragmentation

[15%]
- ii. Write separate **queries** to fragment the relation **horizontally** on the attribute **Account_Type** to split the account details.

[15%]
- iii. Write a **query** to fragment the relation **vertically** to find the **Account_Holder_Name** and **Branch_Code** details.

[10%]
- iv. Write separate **queries** to fragment the relation using **hybrid fragmentation** to find the **Account_ID** for each of the **Account_Type**.

[15%]